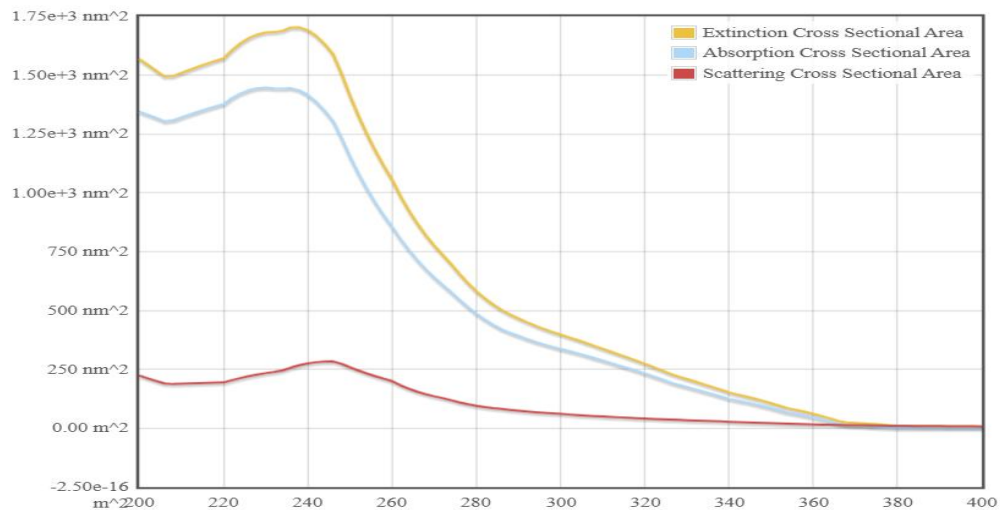


EXPERIMENT 1: Simulation of scattering and absorption of Multi-layer Sphere and Core-Shell Nanoparticle using Mie-Plot tool

Aim

To simulate the scattering and absorption characteristics of multilayer and core-shell nanoparticles using the Mie-Plot tool and study their optical response.

EXP 1: TEST CASE1: Scattering and absorption spectra for a single-layer sphere (core only).



Core/Shell Refractive Indices (RI) only need an input when Core/Shell Type is Other

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Core Type Core RI Core Diameter (nm)

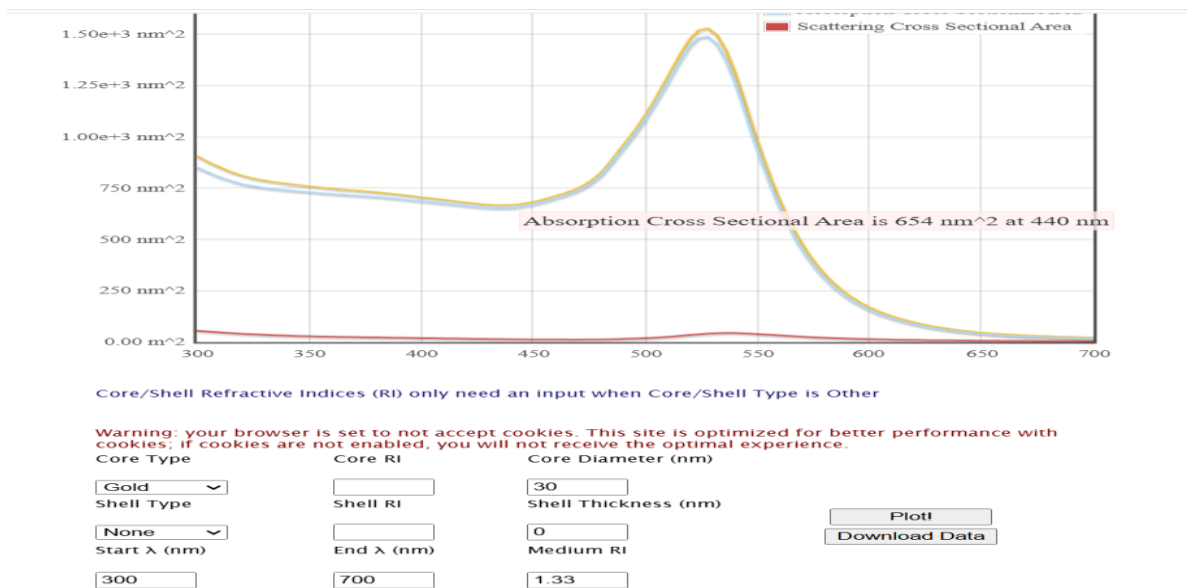
TiO₂ ▼

30

Observation: Extinction and absorption are highest around **240 nm**, while scattering remains low. The curves decrease rapidly as wavelength increases.

Impact: The TiO₂ nanoparticle shows **strong UV absorption**, making it suitable for applications requiring UV light absorption.

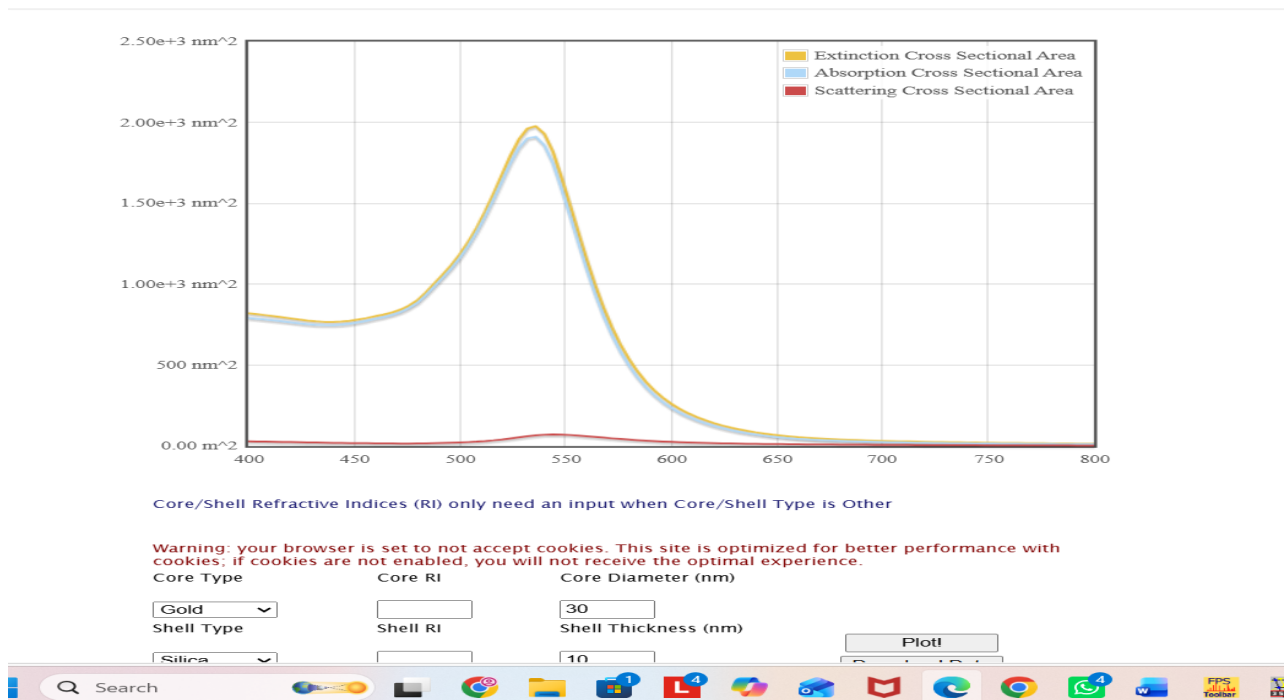
TEST CASE: 2 Simulate the optical response of a gold (Au) nanoparticle in water medium.



Observation: A strong peak occurs around 520–540 nm, with absorption much higher than scattering. The response decreases significantly after the peak.

Impact: The gold nanoparticle exhibits a strong visible-light plasmonic resonance, mainly due to absorption.

TEST CASE 3: Simulate a core-shell nanoparticle with a gold core (30 nm) and silica shell (10 nm).



Observation: A strong extinction and absorption peak occurs around 530–550 nm, while scattering remains comparatively low. The peak is stronger than the response at longer wavelengths.

Impact: The silica shell modifies and enhances the optical response of the gold core, demonstrating that core-shell structures can be used to control nanoparticle optical properties.

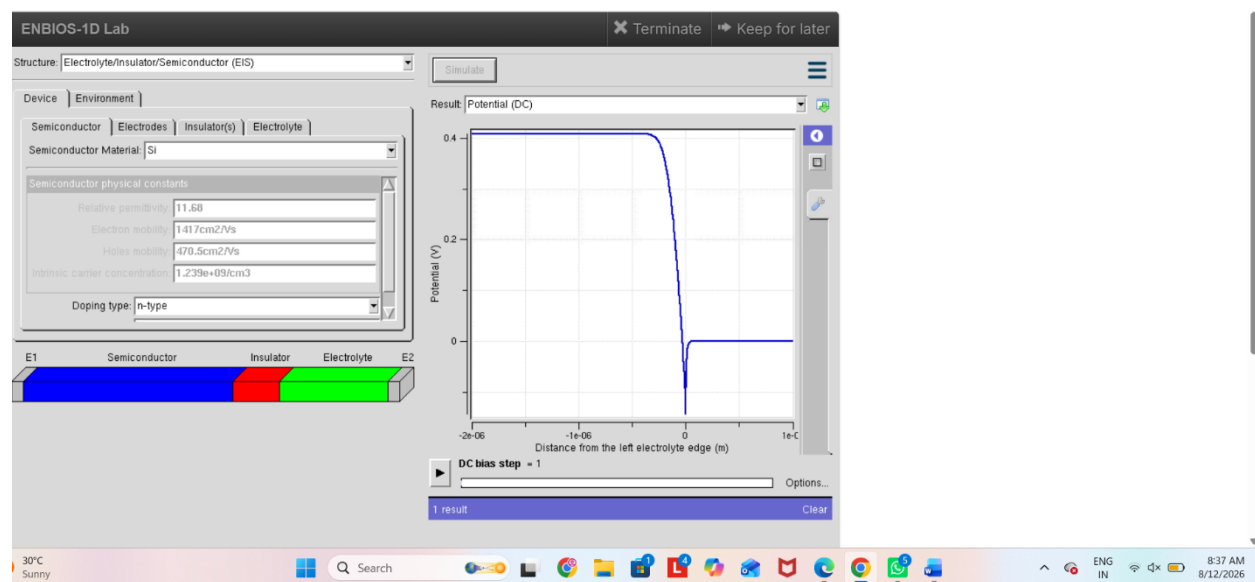
Conclusion

The simulation shows that particle material, size, and shell structure affect scattering and absorption. The core-shell structure modifies the resonance and helps control the nanoparticle's optical properties.

EXPERIMENT 11 Calculate the Potential of a pH SENSOR using NanoHub

Aim

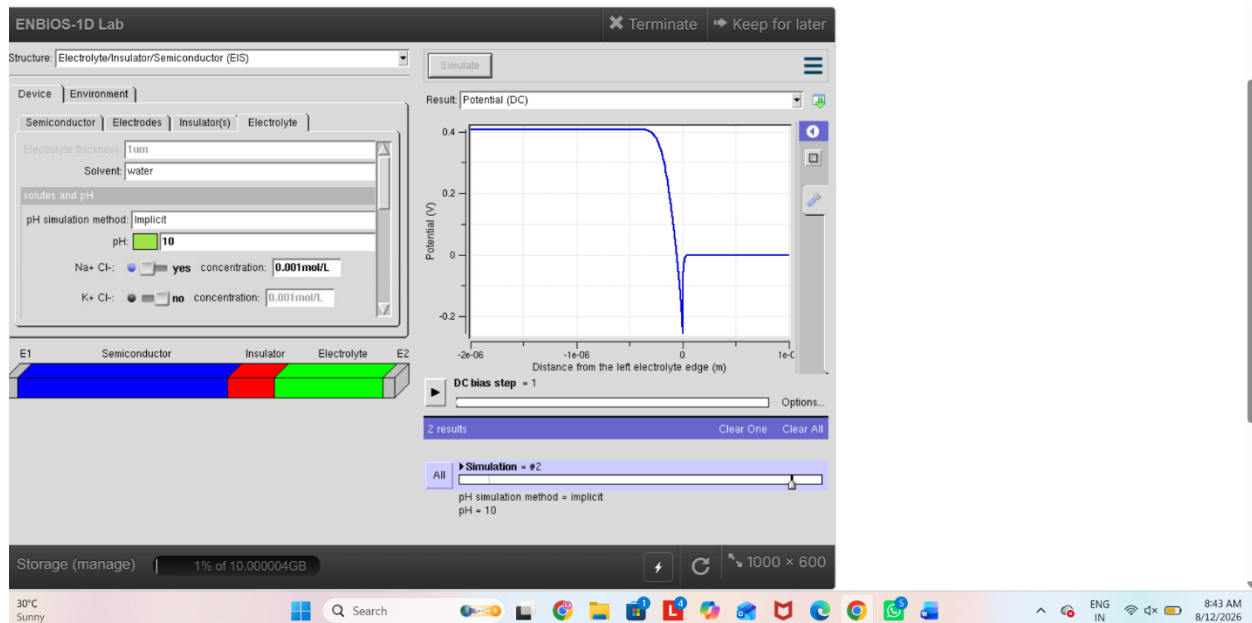
To study the effect of SiO_2 insulator thickness on the potential distribution of an Electrolyte–Insulator–Semiconductor (EIS) structure using the ENBIOS-1D simulation tool.



Observation: The potential remains nearly constant at about **0.4 V** and drops sharply near the semiconductor–insulator–electrolyte interface.

Impact: This sharp potential variation indicates the formation of an **electric field and charge distribution** at the EIS interface.

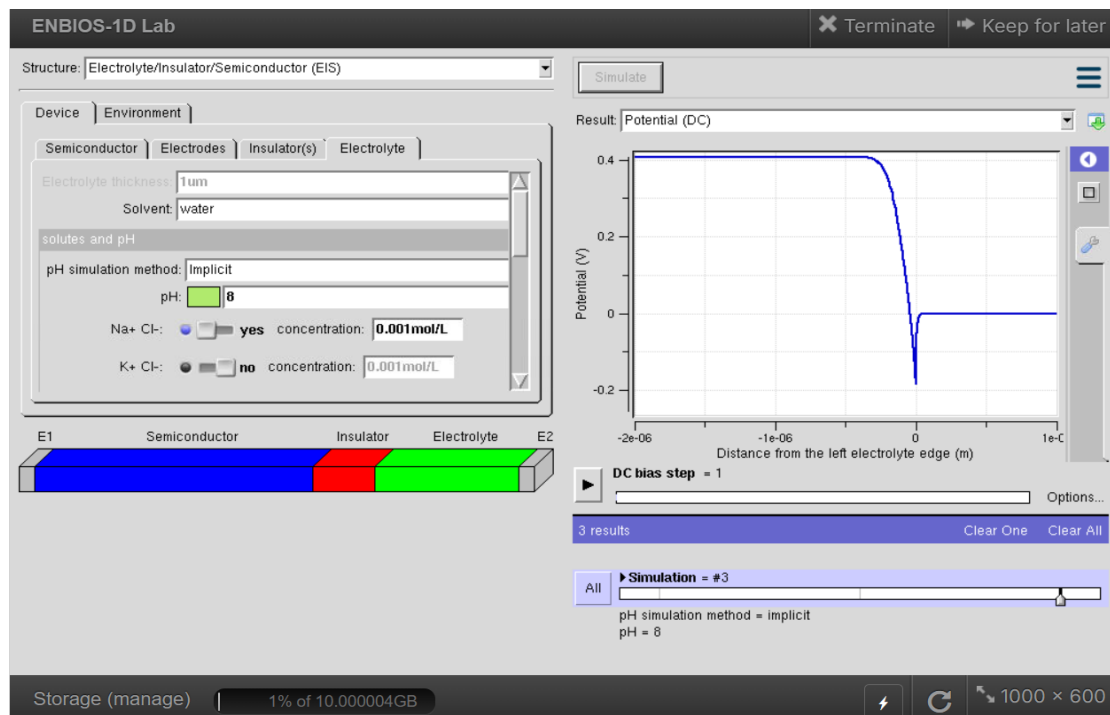
TEST CASE 1: PH = 10



Observation: At pH 10, the potential shows a sharp drop near the interface, reaching a small negative value before returning close to zero.

Impact: The change in pH modifies the **surface charge and electrostatic potential**, showing the sensitivity of the EIS structure to the electrolyte environment.

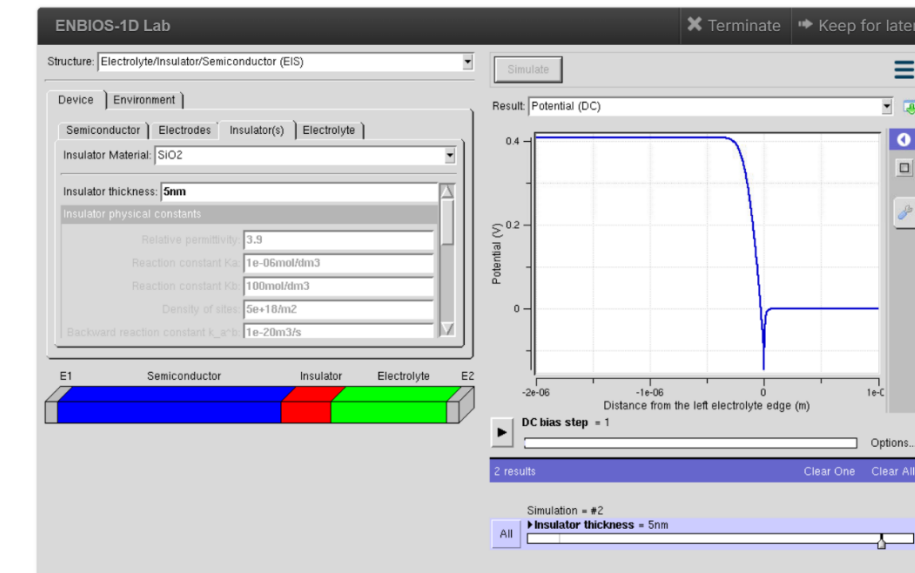
TEST CASE:02 PH =8



Observation: At pH 8, a similar potential drop occurs near the interface, but the minimum potential is different from the pH 10 case.

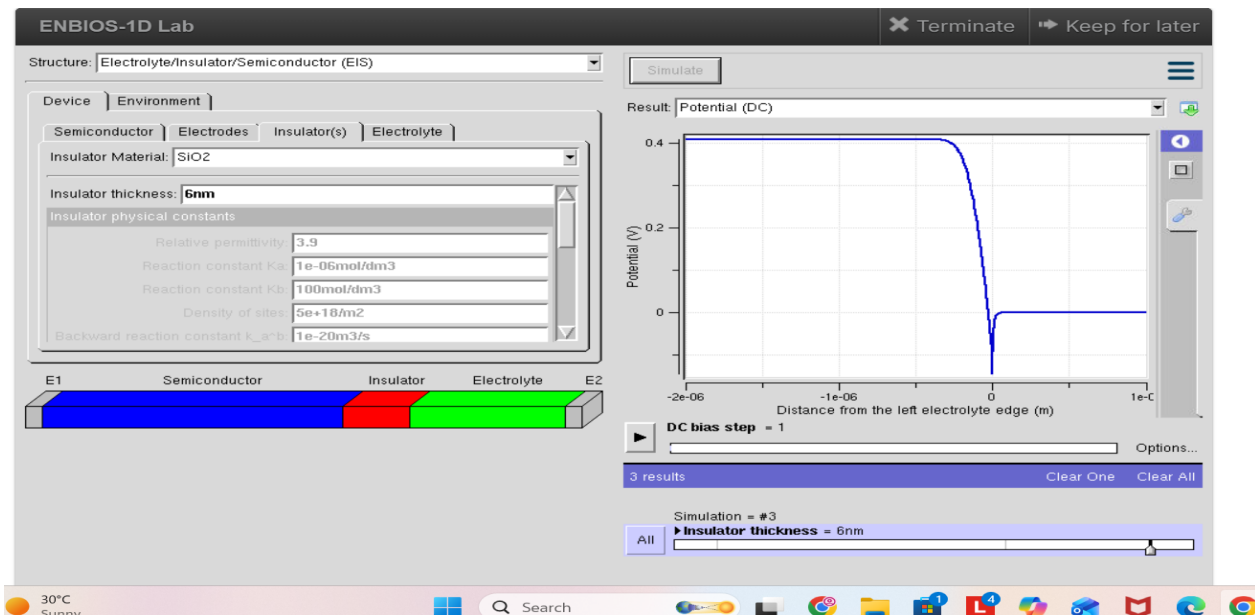
Impact: This confirms that **pH variation changes the surface potential**, demonstrating the working principle of an EIS-based pH/biosensor.

TEST CASE 1: SiO₂ Insulator Thickness = 5 nm



Observation: The potential stays nearly constant at 0.4 V and drops sharply at the semiconductor–insulator interface.

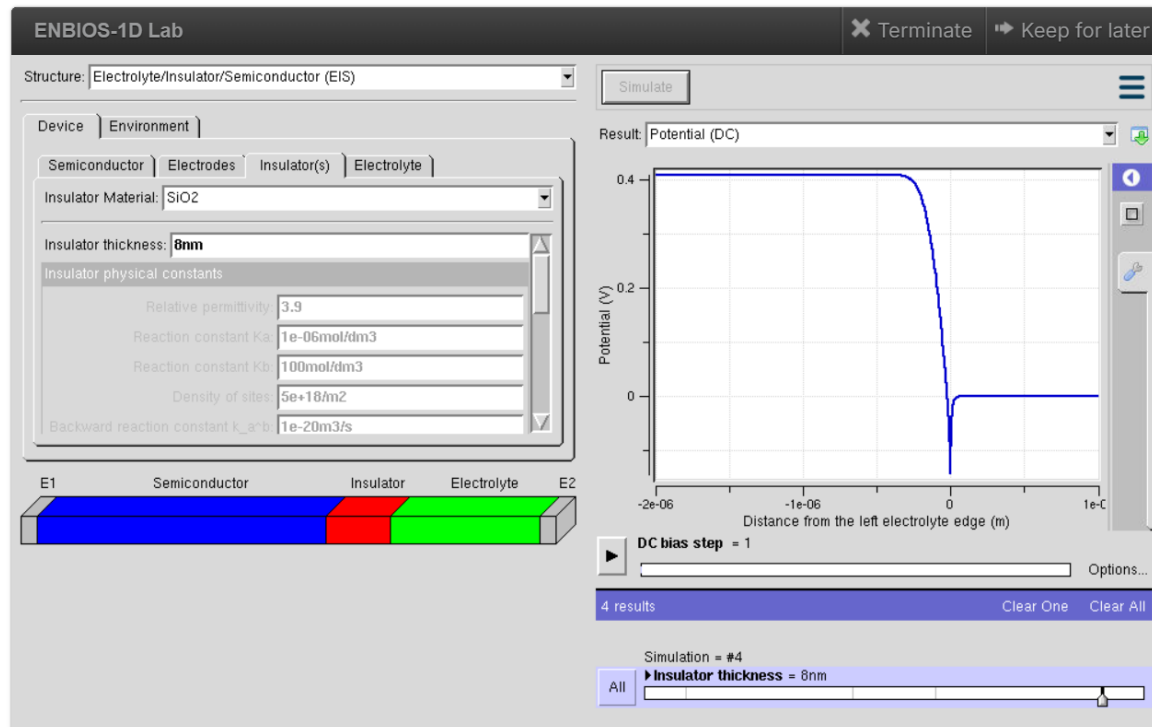
Impact: The 5 nm SiO₂ layer produces a strong potential variation, indicating effective electrostatic control at the EIS interface.



Observation: The potential profile is similar, with a sharp drop near the interface and a slightly different minimum potential.

Impact: Increasing the insulator thickness changes the **potential distribution**, which can influence the sensitivity and electrical response of the EIS sensor.

TEST CASE3: **SiO₂ Insulator Thickness = 8 nm**



Observation: A sharp potential drop is again observed near the interface, while the potential remains almost constant away from the interface.

Impact: The thicker SiO₂ layer modifies the **electric-field distribution and capacitive coupling**, affecting the overall sensor response.

CONCLUSION: The simulation shows that changing the SiO₂ thickness from 5 nm to 8 nm affects the potential distribution near the EIS interface. Hence, insulator thickness is an important parameter for controlling the electrical response and sensitivity of EIS-based biosensors.

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